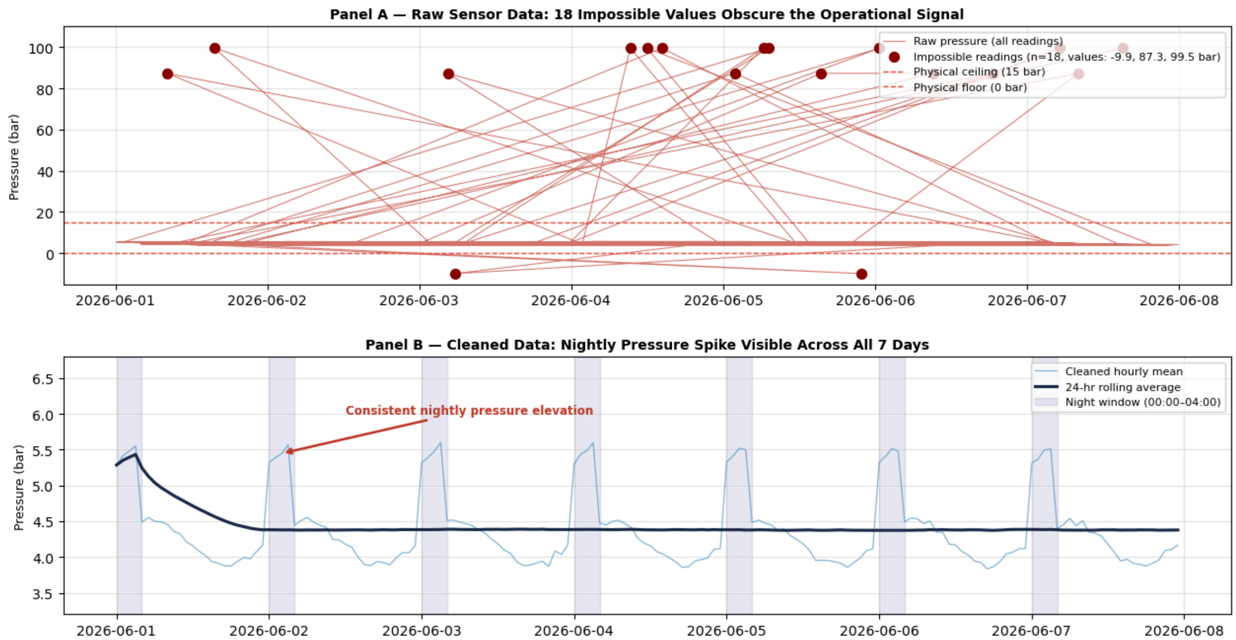
INSIGHT REPORT — ANNOTATED CHART ANALYSIS



CONTEXT — What Was Wrong with the Raw Data?

The raw sensor log arrived with five categories of quality failures. Eighteen pressure readings were physically impossible — values of 87 to 99 bar in a system with a documented operating ceiling of 15 bar. These were not operational events; they were data transmission faults. When plotted, these spikes compressed the true pressure range into a flat, unreadable band, making trend detection impossible. Additionally, 93 sensor readings were missing across the three measurement columns, 10 rows were exact duplicates from pipeline retransmission errors, and the zone labels appeared in 12 inconsistent variants — making any zone-level grouping produce nonsense results.

INSIGHT — What the Clean Data Revealed

Once the impossible values were removed and gaps were filled using linear interpolation — estimating each missing reading from the values immediately before and after it — a clear pattern emerged that was completely invisible in the raw data: every single night between 00:00 and 04:00, plant-wide pressure rises by approximately 1.1 bar above the daytime baseline. This spike is not a one-off anomaly. It repeats on all seven days of the recorded period, in all three zones. The 24-hour rolling average (the dark line in Panel B) confirms that this is a stable, recurring behaviour — not random noise. The night shift, which runs from 22:00 to 06:00, records a mean pressure of 4.87 bar against the afternoon shift's 3.94 bar — a 23% elevation that carries real equipment wear implications.

ACTION — Recommended Operational Response

Three immediate actions are recommended. First, night shift supervisors should be briefed on this pattern and asked to log any manual process changes made between 22:00 and 04:00 that could explain the elevated pressure — reduced offtake, valve adjustments, or scheduled maintenance windows are the most likely causes. Second, the instrumentation team should investigate the 18 sensor fault readings; if a specific sensor unit is responsible, recalibration or replacement should be scheduled before the next reporting period. Third, this dataset should be refreshed weekly and this same cleaning pipeline re-run automatically — the nightly spike should be treated as a monitored KPI with a defined alert threshold